

ENGDATA Notes

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Type of Data

Study of the population is a **census**. Study of a sample of a population is a **survey**.

Data can either be numerical or categorical. Just because the data points are numbers, it doesn't mean they are numerical (e.g. student ID numbers, years).

The **Delphi technique** is when you send rounds of questionnaires to a panel of experts to reach a consensus.

Types of Probability Sampling

1. Simple Random Sampling: Every member has an equal chance of being selected (e.g. lottery, random number generator)
2. Cluster Sampling: The population is divided by clusters, which is often geographical. And entire clusters are randomly selected.
3. Stratified Sampling: The population is divided into subgroups (strata) based on shared characteristics and random samples are taken from each strata.
4. Systematic Sampling: Every n-th member of a group is selected.

The point of probability sampling is that each individual has a known, non-zero chance of being selected; minimizing bias and allowing for generalization.

Types of Non-Probability Sampling

1. Convenience Sampling: Individuals are selected for their availability and accessibility.
2. Voluntary Response Sampling: Individuals select themselves.
3. Purposive/Judgmental Sampling: The researcher uses their expertise to select a sample that is most useful to the purpose of the research.
4. Snowball Sampling: Participants refer other participants.
5. Quota Sampling: Selecting a specific number of individuals from different subgroups.

These methods are faster and easier but risks higher bias. Most often used in exploratory research.

Types of Comparisons

1. Component comparison:
 2. Correlation comparison:
 3. Frequency comparison:
 4. Item comparison:
 5. Time series:
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Probability

Defining Sample Space

Listing/Roster method: enumerating all elements of the sample space.

Property/Set-Builder method: Choosing a property or characteristic common to all sample points and then using this common characteristic to define the sample space.

Tree Diagram: Uses a tree data structure to express sequential order of events of an experiment in chronological order.

An **event** is a subset of a sample space.

A **sample point** is a specific outcome from the **sample space**.

A **simple event** is an event which only has one outcome.

A **compound event** is an event which consists of more than one outcome and may be decomposed into simple events.

Operations on Events

Union: $A \cup B$, set of all events in A or B.

Intersection: $A \cap B$. Set of all events in A and B

Complement: A' represents the set of all events not found in A.

S represents the sample space. So S' represents all events not in the sample space.

$A \& B = \text{NULL}$ is **mutually exclusive** or **disjointed**.

Counting Principles and Techniques

Counting Theory

A set of techniques used for identifying the total number of possibilities of a statistical experiment without enumerating the elements of the sample space.

Multiplication Rule:

$$n_T = n_1 \times n_2 \times \dots \times n_k$$

Where n is an outcome.

Addition Rule: If k outcomes are mutually exclusive, and outcome 1 can be performed in m_1 ways, outcome 2 can be performed in m_2 ways, and so on for k outcomes, then the total number of outcomes is given by:

$$m_T = m_1 + m_2 + \dots + m_k$$

Simple Permutation

There is an ordered arrangement of **distinct objects**.

The formula:

$${}_nP_r = \frac{n!}{(n-r)!}$$

Simple Combination

An arrangement of distinct objects without regard to order.

$${}_nC_r = \frac{n!}{(n-r)! * r!}$$

Special types of permutation

6.4.1 Circular Permutation

Arrangement of objects in a circle.

The formula is:

$$P_{nc} = (n-1)!$$

6.4.2 Arrangement of Similar Objects

To find the number of permutations of n "objects" all taken at the same time.

- n_1 are objects of type 1.
- n_2 are objects of type 2.
- n_k are objects of type k .

- n represents the total objects.

$$\sum_{i=1}^k n_i = n$$

This translates to:

$$\frac{n!}{n_1!n_2!n_3!\dots n_k!}$$

6.4.3 Partitioning of n objects into cells

To find the number of permutations of n "objects" all taken at the same time.

- n_1 are the objects placed in cell 1.
- n_2 are the objects placed in cell 2.
- n_k are the objects placed in cell k .

Very similar to the previous type of permutation.

$$\sum_{i=1}^k n_i = n$$

This translates to:

$$\frac{n!}{n_1!n_2!n_3!\dots n_k!}$$

Axioms of Probability

Axiom 1: $0 \leq P(A) \leq 1$.

Axiom 2: $P(A) = 1$ if A is a certain or sure event.

Axiom 3: $P(A) = 0$ if A is an impossible event.

Axiom 4: If the probability of events in the sample space is added, the sum is always 1.

$$\sum_{i=1}^n P(E) = 1$$

Ways of measuring probability

Classical Concept

The probabilities of events can be determined without gathering data.

$$P(S) = \frac{n(A)}{N}$$

Where $n(A)$ is the subset of the sample space and N is the total sample space.

Relative Frequency

Uses historical data or survey results to determine probabilities.

$$P(A) = \frac{f_A}{f_T}$$

Where f_A is the frequency of occurrence of event A and f_T is the total number of observations.

Subjective Assessment

An expert in the field provides an educated guess of the probability.

Probability Rules
